

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - Scenario Based Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Scenario Based Questions	Assessment:	ASSESSMENT TOOL 1 - Scenario Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze sample data using frequentist statistical inference, point estimation, and confidence interval techniques to estimate population parameters and support industry-based decision making. (BL4)		
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Section / Batch:	2024	Date:	13.08.2026

Code of Conduct

I Jeya Harshini R (192424051) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jeya Harshini R

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question 1 – Point Estimate

Question

A food delivery company wants to know the average delivery time in Chennai. It randomly selects 100 deliveries and obtains an average delivery time of 32 minutes. What population parameter is being estimated, and what is the point estimate?

Answer

Step 1: Identify the Population and Sample

Population: All food deliveries made by the company in Chennai.

Sample: 100 randomly selected deliveries.

$$n = 100$$

Step 2: Identify the Population Parameter

The population parameter being estimated is the **population mean delivery time**.

It is denoted by:

$$\mu$$

where (μ) represents the true average delivery time of all deliveries in Chennai.

Step 3: Calculate the Point Estimate

The sample mean is used as the point estimate of the population mean.

Formula:

$$\hat{\mu} = \bar{x}$$

Given:

$$\bar{x} = 32 \text{ minutes}$$

Therefore:

$$\hat{\mu} = 32 \text{ minutes}$$

Step 4: Interpretation

Based on the sample of 100 deliveries, the company estimates that the average delivery time of all deliveries in Chennai is **32 minutes**.

Final Answer

Population Parameter:

μ = Population Mean Delivery Time

Point Estimate:

$\hat{\mu} = 32$ minutes

Therefore, the population parameter being estimated is the **average delivery time of all deliveries in Chennai**, and the **point estimate is 32 minutes**.

Question 2 – Point Estimate

Question

An e-commerce website wants to estimate the percentage of customers who purchase a product after viewing its advertisement. From **800 randomly selected visitors, 176 made a purchase**. Determine the point estimate of the population proportion.

Answer

Step 1: Identify the Population and Sample

Population: All visitors who view the advertisement on the e-commerce website.

Sample: 800 randomly selected visitors.

$n = 800$

Number of visitors who made a purchase:

$x = 176$

Step 2: Identify the Population Parameter

The population parameter being estimated is the **population proportion of customers who make a purchase after viewing the advertisement**.

It is represented by:

$$p$$

Step 3: Calculate the Point Estimate

The point estimate of a population proportion is the **sample proportion**.

Formula:

$$\hat{p} = \frac{x}{n}$$

Substituting the values:

$$\hat{p} = \frac{176}{800}$$
$$\hat{p} = 0.22$$

Converting to percentage:

$$0.22 \times 100 = 22\%$$

Step 4: Interpretation

Based on the sample data, approximately **22% of visitors who viewed the advertisement made a purchase**.

This estimate can be used by the e-commerce company to evaluate the effectiveness of its advertising campaign.

Final Answer

Population Parameter:

p = Population Proportion of Customers Making a Purchase

Point Estimate:

$$\hat{p} = \frac{176}{800} = 0.22$$

$$\hat{p} = 22\%$$

Therefore, the **point estimate of the population proportion is 0.22 or 22%**.

Question 3 – Frequentist Approach

Question

A bank wants to estimate the average monthly expenditure of its credit-card customers. A random sample of **150 customers** has a mean expenditure of **₹24,500**. Explain how the sample mean can be used as a frequentist estimate of the population mean.

Answer

Step 1: Identify the Population and Sample

Population: All credit-card customers of the bank.

Sample: 150 randomly selected credit-card customers.

$$n = 150$$

Sample mean expenditure:

$$\bar{x} = ₹24,500$$

Step 2: Identify the Population Parameter

The population parameter being estimated is the **population mean monthly expenditure** of all credit-card customers.

It is represented by:

$$\mu$$

Step 3: Frequentist Estimation

According to the **frequentist approach**, the population parameter (μ) is fixed but unknown.

A random sample is selected from the population, and the **sample mean** ($\bar{x}$) is used as a point estimate of the population mean.

Formula:

$$\hat{\mu} = \bar{x}$$

Substituting the given value:

$$\hat{\mu} = ₹24,500$$

Step 4: Explanation

The frequentist approach assumes that if many random samples of 150 customers were selected, each sample would produce a different sample mean.

These sample means would form a sampling distribution centered around the true population mean.

Therefore, the sample mean of **₹24,500** is considered the best estimate of the population mean monthly expenditure.

Step 5: Interpretation

Based on the sample data, the bank estimates that the average monthly expenditure of all its credit-card customers is approximately **₹24,500**.

This estimate can help the bank understand customer spending behavior and design suitable financial products and services.

Final Answer

Population Parameter:

$$\mu = \text{Population Mean Monthly Expenditure}$$

Sample Mean:

$$\bar{x} = ₹24,500$$

Frequentist Estimate:

$$\hat{\mu} = ₹24,500$$

Therefore, under the **frequentist approach**, the sample mean **₹24,500** is used as the point estimate of the population mean monthly expenditure of all credit-card customers.

Question 4 – Point Estimate

Question

A streaming platform selects **500 subscribers** to estimate the percentage of users who watch movies every day. If **320 users** report watching movies daily, calculate the point estimate and explain its interpretation.

Answer

Step 1: Identify the Population and Sample

Population: All subscribers of the streaming platform.

Sample: 500 randomly selected subscribers.

$$n = 500$$

Number of subscribers who watch movies daily:

$$x = 320$$

Step 2: Identify the Population Parameter

The population parameter being estimated is the **population proportion of subscribers who watch movies every day**.

It is represented by:

$$p$$

Step 3: Calculate the Point Estimate

The point estimate of a population proportion is the **sample proportion**.

Formula:

$$\hat{p} = \frac{x}{n}$$

Substituting the values:

$$\hat{p} = \frac{320}{500}$$

$$\hat{p} = 0.64$$

Converting to percentage:

$$0.64 \times 100 = 64\%$$

Step 4: Interpretation

The sample indicates that **64% of subscribers watch movies every day**.

This estimate suggests that approximately **64 out of every 100 subscribers** on the streaming platform are daily movie viewers.

The company can use this information to plan content recommendations, advertising strategies, and subscription services.

Final Answer

Population Parameter:

p = Population Proportion of Daily Movie Viewers

Point Estimate:

$$\hat{p} = \frac{320}{500}$$

$$\hat{p} = 0.64$$

$$\hat{p} = 64\%$$

Therefore, the **point estimate of the population proportion is 0.64 or 64%**, indicating that approximately 64% of subscribers watch movies every day.

Question 5 – Confidence Interval

Question

A manufacturing company randomly selects **64 smartphones** and finds that their average battery life is **18.5 hours** with a standard deviation of **2 hours**. Construct a **95% confidence interval** for the average battery life.

Answer

Step 1: Identify the Population and Sample

Population: All smartphones produced by the manufacturing company.

Sample: 64 randomly selected smartphones.

$$n = 64$$

Sample mean battery life:

$$\bar{x} = 18.5 \text{ hours}$$

Sample standard deviation:

$$s = 2 \text{ hours}$$

Step 2: Identify the Population Parameter

The parameter being estimated is the **population mean battery life**.

It is represented by:

$$\mu$$

Step 3: Determine the Confidence Level

Given confidence level:

$$95\%$$

For a 95% confidence interval:

$$z_{\alpha/2} = 1.96$$

Step 4: Calculate the Standard Error

Formula:

$$SE = \frac{s}{\sqrt{n}}$$

Substituting values:

$$SE = \frac{2}{\sqrt{64}}$$

$$SE = \frac{2}{8}$$

$$SE = 0.25$$

Step 5: Calculate the Margin of Error

Formula:

$$ME = z \left(\frac{s}{\sqrt{n}} \right)$$

$$ME = 1.96(0.25)$$

$$ME = 0.49$$

Step 6: Construct the Confidence Interval

Formula:

$$CI = \bar{x} \pm ME$$

Substituting values:

$$CI = 18.5 \pm 0.49$$

Lower Limit:

$$18.5 - 0.49 = 18.01$$

Upper Limit:

$$18.5 + 0.49 = 18.99$$

Step 7: Confidence Interval

$$(18.01, 18.99)$$

or

18.01 hours to 18.99 hours

Step 8: Interpretation

We are **95% confident** that the true average battery life of all smartphones produced by the company lies between **18.01 hours and 18.99 hours**.

This interval provides a range of plausible values for the population mean battery life.

Final Answer

Population Parameter:

μ = Population Mean Battery Life

95% Confidence Interval:

$$CI = \bar{x} \pm z \left(\frac{s}{\sqrt{n}} \right)$$

$$CI = 18.5 \pm 1.96 \left(\frac{2}{8} \right)$$

$$CI = 18.5 \pm 0.49$$

$$CI = (18.01, 18.99)$$

Therefore, the **95% confidence interval for the average battery life is 18.01 hours to 18.99 hours.**